

Tharun Vaithya Devaraj

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LinkedIn - [tharun-vaithya-devaraj](#) | **Portfolio** - [tharun-vaithya.github.io/portfolio](#)

Summary

Results-oriented engineering professional with experience in operations management, process optimization, and workflow efficiency. Proficient in implementing continuous improvement initiatives, ensuring regulatory compliance, and leveraging data-driven solutions to enhance manufacturing performance and productivity. Seeking opportunities to apply technical expertise and problem-solving skills to deliver impactful results.

Education

MS in Mechanical Engineering, San Jose State University

- **Coursework:** Advanced Manufacturing, Mechanical Design and Simulation, Engineering Management, Computational Engineering/CAE

BE in Mechanical Engineering, St. Joseph's Institute of Technology

- **Coursework:** Manufacturing Basics, Mechanical Fundamentals, Design and Analysis, Mechanics of Materials

Work Experience

Trainee, Vestas Wind Systems

July 2023 – July 2024

- Optimized Manufacturing Execution Systems (MES) to enhance production efficiency by 25%, ensuring seamless integration with ERP systems for real-time data visibility
- Led the successful implementation and upgrades of MES projects, achieving 100% on-time delivery
- Diagnosed and resolved system issues, reducing downtime by 40% through effective troubleshooting and proactive maintenance
- Developed training materials and conducted sessions for end-users, resulting in a 20% increase in user adoption and system utilization.

Intern, Flowwell Castings

Feb 2022 – Mar 2022

- Performed technical analysis using engineering software to conduct data analysis, simulations, and modeling, leading to a 15% improvement in project outcomes in a short span
- Assisted in the creation and updating technical drawings and project documents, ensuring accuracy and compliance with industry standards
- Tested and evaluated prototypes with Coordinate Measuring Machines (CMM), achieving precise measurements and comprehensive data analysis.

Project Experience

Study of Polymer Composite

- Analyzed the viscoelastic and dynamic properties of jute and pineapple fiber polymer composites to assess mechanical performance
- Utilized Dynamic Mechanical Analysis (DMA) for precise material evaluation
- Developed reports and presentations to communicate findings and recommend optimizations

Effi-cycle (Design of Hybrid Vehicle)

- Led the Powertrain Design Team to achieve a national ranking of 9, utilizing design software such as SolidWorks and CATIA to develop detailed plans, prototypes, and simulations
- Performed FEA and CFD simulations using ANSYS, integrating results to enhance design performance.

Skills

- **Core skills:** Operations Management, Manufacturing Process Improvement, Engineering Management, Product Support, MES, MOM, EAM, Six Sigma, Thermal Analysis, FEA, CAE, Process Improvement, Design for Manufacturability, Failure Analysis, SQL, Project management, Power BI, Root Cause Analysis
- **Technical Software:** QDA, Apriso, infor EAM, aPriori, Windchill, Solidworks, CATIA, CREO, AutoCAD, Ansys, MATLAB, Minitab
- **Softskills:** Communication, Teamwork, Problem-solving, Adaptability, Leadership, Critical Thinking